WORKSHEET

Chapter 6: Finding indefinite integrals 2

1 Find each of the following indefinite integrals.

a $\int (3x^2 - 6x + 9)dx$

b $\int (2x^2 - 5x + 3)dx$

c $\int (15x^4 - 20x^3 + 8x)dx$

d $\int (5x^4 + 20x^3 - 9x^2 - 9)dx$

e $\int \sqrt{x}(1 - x)\,dx$

f $\int (-3x^{-4} + 6x^{-3} - 5x^{-2})dx$

g $\int \left(\frac{24}{m^3} - \frac{8}{m^5} - \frac{12}{m^7}\right)dx$

h $\int (3x + 1)(2 - x)dx$

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i $\int \frac{x^3 + 5x^2 - 4}{x^2}\, dx$

j $\int x(x-1)^2\, dx$

2 Find each of the following.

a $\int (e^{5x} + 7x)dx$

b $\int (e^{x} + e^{-x})dx$

c $\int \left(e^{2x} - e^{\frac{x}{3}}\right)dx$

d $\int \frac{e^{2x} + 2}{e^{x}}\, dx$

e $\int (3e^{\frac{x}{3}} - 3e^{\frac{x}{5}})dx$

f $\int \frac{e^{x} - e^{-x}}{3}\, dx$

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g $\int (e^{3x} + 1)dx$

h $\int (e^{x} - 1)^2 dx$

3 Find the indefinite integral of each of the following.

a $-\cos(x) - \sin(x)$

b $\sin(3x) + \cos(2x)$

c $\cos(4x) - \sin(2x)$

d $\sin\left(\frac{x}{2}\right) - \cos(4x)$

e $\cos\left(\frac{x}{3}\right) - 4\sin(2x)$

f $5x + 2\cos(x)$

g $4\cos(4x) - \frac{1}{3}\sin(2x)$

h $e^{2x} - 4\sin(8x) + 3$

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4 Find the indefinite integral of each of the following.

a $3 + 4\sin(8x) + 3e^{5x}$

b $e^{4x} + \sin(3x) + 2x^3$

c $6x^2 - 4\cos(2x) + 9e^{3x}$

d $4x^2 + 8\cos(2x) - e^{-x}$

e $2\sin\left(\frac{x}{3}\right) + \frac{1}{4}e^{\frac{x}{2}} - (2x-1)^4$

f $5\sin\left(\frac{x}{2}\right) - 4\cos\left(\frac{x}{3}\right) + e^{\frac{-x}{4}}$

g $\sqrt{x} + 4x + 2\sin\left(\frac{\pi x}{3}\right) + 9$

h $e^{3x} + 4\cos(2x) - (4x+3)^3$

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Answers

1 **a** $x^3 - 3x^2 + 9x + c$

b $\frac{2}{3}x^3 - \frac{5}{2}x^2 - 3x + c$

c $3x^5 - x^3 + 4x^2 + c$

d $x^5 + 5x^4 - 3x^3 - 9x + c$

e $2\sqrt{x^3}\left(\frac{1}{3} - \frac{1}{5}x\right) + c$

f $\frac{1}{x^3} - \frac{3}{x^2} + \frac{5}{x} + c$

g $\frac{-12}{x^2} + \frac{2}{x^4} + \frac{2}{x^6} + c$

h $2x - \frac{5}{2}x^2 - x^3 + c$

i $\frac{1}{2}x^2 + 5x + \frac{4}{x} + c$

j $\frac{1}{4}x^4 - \frac{2}{3}x^3 + \frac{1}{2}x^2 + c$

2 **a** $\frac{1}{5}e^{5x} + \frac{7}{2}x^2 + c$

b $e^x - e^{-x} + c$

c $\frac{1}{2}e^{2x} - 3e^{\frac{x}{3}} + c$

d $e^x - 2e^{-x} + c$

e $9e^{\frac{x}{3}} - 15e^{\frac{x}{5}} + c$

f $\frac{e^x + e^{-x}}{3} + c$

g $\frac{1}{3}e^{3x} + x + c$

h $\frac{1}{2}e^{2x} - 2e^x + x + c$

3 **a** $\sin(x) + \cos(x) + c$

b $\frac{1}{2}\sin(2x) - \frac{1}{3}\cos(3x) + c$

c $\frac{1}{4}\sin(4x) + \frac{1}{2}\cos(2x) + c$

d $-2\cos\left(\frac{x}{2}\right) - \frac{1}{4}\sin(4x) + c$

e $3\sin\left(\frac{x}{3}\right) + 2\cos(2x) + c$

f $\frac{5}{2}x^2 + 2\sin(x) + c$

g $\sin(4x) + \frac{1}{6}\cos(2x) + c$

h $\frac{1}{2}e^{2x} + \frac{1}{2}\cos(8x) + 3x + c$

4 **a** $\frac{3}{5}e^{5x} - \frac{1}{2}\cos(8x) - 3x + c$

b $\frac{1}{4}e^{4x} - \frac{1}{3}\cos(3x) + \frac{1}{2}x^4 + c$

c $2x^3 - 2\sin(2x) + 3e^{3x} + c$

d $\frac{4}{3}x^3 + 4\sin(2x) + e^{-x} + c$

e $-6\cos\left(\frac{x}{3}\right) + \frac{1}{2}e^{\frac{x}{2}} - \frac{1}{8}(3x-1)^5 + c$

f $-10\cos\left(\frac{x}{2}\right) - 12\sin\left(\frac{x}{3}\right) - 4e^{-4x} + c$

g $\frac{2}{3}x^{\frac{3}{2}} + 2x^2 - \frac{6}{\pi}\cos\left(\frac{\pi x}{3}\right) + 9x + c$

h $\frac{1}{3}e^{3x} + 2\sin(2x) - \frac{1}{16}(4x+3)^4 + c$

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